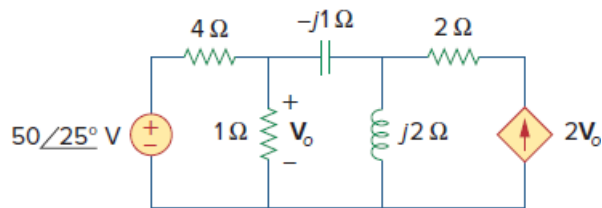


Problem

For the circuit in Fig. 11.76, find the average, reactive, and complex power delivered by the dependent current source.

Figure 11.76:



Step-by-step solution

Step 1 of 10

Refer to Figure 11.76 in the text book.

Consider the circuit diagram shown in Figure 1.

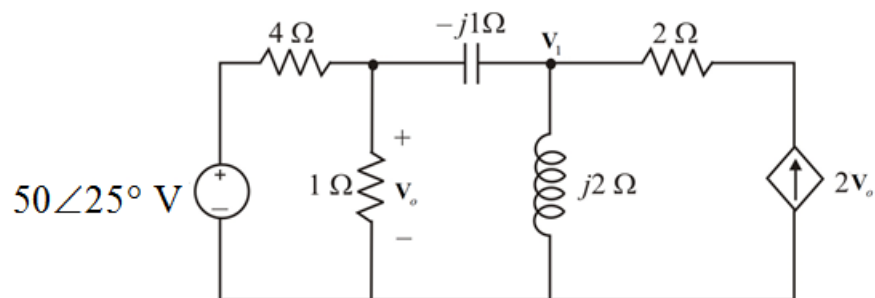


Figure 1

Step 2 of 10

Apply Kirchhoff's Voltage Law at node 2.

$$\frac{V_1 - V_o}{-j1} + \frac{V_1}{j2} - 2V_o = 0$$

$$V_1 \left(\frac{1}{-j1} + \frac{1}{j2} \right) - V_o \left(2 + \frac{1}{-j1} \right) = 0$$

$$V_1 (j1 - j0.5) - V_o (2 + j1) = 0$$

$$V_1 (j0.5) = V_o (2 + j1)$$

$$V_1 = V_o \left(\frac{2 + j1}{j0.5} \right)$$

$$V_1 = V_o (2 - j4) \dots\dots (1)$$

Therefore, the expression for the voltage V_1 is $V_o (2 - j4)$.

Step 3 of 10

Apply Kirchhoff's Voltage Law in loop-1.

$$\frac{V_o - V_1}{-j1} + \frac{V_o}{1} + \frac{V_o - 50\angle 25^\circ}{4} = 0$$

Substitute $V_o(2 - j4)$ for V_1 .

$$\frac{V_o - V_o(2 - j4)}{-j1} + \frac{V_o}{1} + \frac{V_o - 50\angle 25^\circ}{4} = 0$$

$$V_o \left(\frac{1 - 2 + j4}{-j1} + 1 + \frac{1}{4} \right) - \frac{50\angle 25^\circ}{4} = 0$$

$$V_o \left(\frac{-1 + j4}{-j1} + 1 + 0.25 \right) - 12.5\angle 25^\circ = 0$$

$$V_o(-4 - j1 + 1 + 0.25) = 12.5\angle 25^\circ$$

$$V_o(-2.75 - j1) = 12.5\angle 25^\circ$$

$$V_o(2.926\angle -160^\circ) = 12.5\angle 25^\circ$$

$$V_o = \frac{12.5\angle 25^\circ}{2.926\angle -160^\circ}$$

$$V_o = 4.272\angle -175^\circ \text{ V}$$

Therefore, the value of the voltage V_o is $4.272\angle -175^\circ \text{ V}$.

[Comments \(1\)](#)



Anonymous

Apply Kirchhoff's "current" law at node 1; is what it should say because that's what is being applied.

Step 4 of 10

Recall equation (1).

$$V_1 = V_o(2 - j4)$$

Substitute $4.272\angle -175^\circ \text{ V}$ for V_o .

$$\begin{aligned} V_1 &= (4.272\angle -175^\circ)(2 - j4) \\ &= (4.272\angle -175^\circ)(4.47\angle -63.44^\circ) \\ &= 19.1\angle 121.56^\circ \text{ V} \end{aligned}$$

Therefore, the value of the voltage V_1 is $19.1\angle 121.56^\circ \text{ V}$.

Step 5 of 10

Calculate the value of the dependent current source.

$$I = 2V_o$$

Substitute $4.272\angle -175^\circ \text{ V}$ for V_o .

$$\begin{aligned} I &= 2(4.272\angle -175^\circ) \\ &= 8.544\angle -175^\circ \text{ A} \end{aligned}$$

Therefore, the value of the dependent current source is $8.544\angle -175^\circ \text{ A}$.

Step 6 of 10

Calculate the value of the voltage across the dependent current source.

$$\begin{aligned}V &= V_1 - (-2V_o)(2) \\ &= V_1 + 4V_o\end{aligned}$$

Substitute $4.272\angle -175^\circ \text{ V}$ for V_o and $19.1\angle 121.56^\circ \text{ V}$ for V_1 .

$$\begin{aligned}V &= 19.1\angle 121.56^\circ + 4(4.272\angle -175^\circ) \\ &= 19.1\angle 121.56^\circ + 17.09\angle -175^\circ \\ &= 30.8\angle 151.31^\circ \text{ V}\end{aligned}$$

Therefore, the value of the voltage across the dependent current source is

$$30.8\angle 151.31^\circ \text{ V}.$$

Step 7 of 10

Calculate the complex power delivered by the dependent current source.

$$S = V(I)^*$$

Substitute $8.544\angle -175^\circ \text{ A}$ for I and $30.8\angle 151.31^\circ \text{ V}$ for V .

$$\begin{aligned}S &= (30.8\angle 151.31^\circ)(8.544\angle -175^\circ)^* \\ &= (30.8\angle 151.31^\circ)(8.544\angle 175^\circ) \\ &= 263.16\angle -33.69^\circ \text{ VA} \\ &= 218.96 - j145.97 \text{ VA}\end{aligned}$$

Therefore, the value of the complex power delivered by the dependent current source is

$$218.96 - j145.97 \text{ VA}.$$

[Comments \(2\)](#)



Anonymous

solution is incorrect. I've come across many incorrect solutions with regard to this text. It makes me wonder what I'm paying for...?



Anonymous

Your formula is incorrect. $S = 1/2 VI^*$ is the proper form.

Step 8 of 10

Compare $218.96 - j145.97 \text{ VA}$ with $P + jQ$ where P represents the real power and Q represents the reactive power.

Therefore, the value of the real and reactive powers are 218.96 W and -145.97 VAR respectively.

Step 9 of 10

MATLAB calculations:

Use equations (1), (2), (3), (4), (5) and (6) in MATLAB to solve the problem.

Write the following instructions in MATLAB m-file and save the program.

```
clc
```

```
syms Vo
```

```
Vo = solve(((Vo-Vo*(2-4i))/-1j)+(Vo/1)+(Vo-45.315-21.231i)/4)% equation to solve voltage Vo
```

```
Vo = double(Vo)%instruction for decimal point representation
```

```
V1 = Vo*(2-4i)% equation for voltage V1
```

```
I = 2*Vo% value of the dependent current source
```

```
V = V1+4*Vo% voltage across the dependent current source
```

```
S = V*conj(I)%complex power delivered by the dependent current source
```

```
P = real(S)% real power delivered by the dependent current source
```

```
Q = imag(S)%reactive power delivered by the dependent current source
```

Step 10 of 10

Run the program and view the results in the command window.

Vo =

- (52281*i)/137000 - 583389/137000

Vo =

-4.2583 - 0.3816i

V1 =

-10.0431 +16.2700i

I =

-8.5166 - 0.7632i

V =

-27.0763 +14.7436i

S =

2.1935e+002 -1.4623e+002i

P =

219.3464

Q =

-146.2309

Therefore, the values of the complex power, real power and reactive power are

$219.35 - j146.23 \text{ VA}$, 219.35 W and 146.23 VAR respectively and are approximately equal to the hand calculated value.